

CSA1217-ARTIFICIAL INTELLIGENCE

CO1- ASSESSMENT TOOL 3

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Question 1: Smart Home Automation System

Introduction

A smart home automation system is an intelligent system that automatically manages home appliances such as lights, fans, air conditioners, heaters, and security devices. It continuously monitors the surrounding environment using sensors and responds according to user preferences and environmental conditions. Over time, the system learns user habits and provides personalized services while improving comfort, energy efficiency, and security.

Simple Reflex Agent

Acts only on the current percept using condition-action rules. Example: If motion is detected at night, the system turns on the hallway light immediately.

Model-Based Agent

Maintains an internal model of the home. It remembers whether doors are locked, rooms are occupied, and previous sensor readings to make better decisions.

Goal-Based Agent

Works towards achieving predefined goals such as maintaining room temperature at 24°C, reducing energy consumption, or ensuring all doors are locked before bedtime.

Utility-Based Agent

Selects the action that provides the highest overall benefit by balancing user comfort, electricity usage, security, and appliance health.

Comparison of Intelligent Agents

Agent Type	Advantages	Limitations
Simple Reflex	Fast response, easy to implement	Cannot learn or remember past events
Model-Based	Handles partially observable environments	Requires internal memory
Goal-Based	Focuses on achieving objectives	Needs planning
Utility-Based	Chooses the most beneficial action	More computationally expensive

Most Suitable Approach

A hybrid approach combining Model-Based and Utility-Based agents is the most effective. The model-based component maintains information about occupants and devices, while the utility-based component selects actions that maximize comfort, minimize energy consumption, and improve security. Machine learning further helps the system adapt to changing user preferences over time.

Conclusion

Therefore, a hybrid intelligent agent provides accurate decision-making, personalized automation, reduced electricity consumption, and improved safety, making it ideal for smart home environments.

Question 2: Robot Navigation in an Unknown Maze

Introduction

A robot operating in an unknown maze must explore the environment, avoid obstacles, and locate the exit without prior knowledge of the map. Search algorithms help the robot systematically explore possible paths and find an optimal solution.

Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the lowest cumulative path cost first. It guarantees the optimal path when all path costs are non-negative. UCS is suitable when movement costs vary between locations.

Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs depth-limited searches while gradually increasing the depth limit. It combines the low memory usage of Depth-First Search with the completeness of Breadth-First Search.

Comparison

Feature	UCS	IDS
Optimal	Yes	Only for equal step cost
Complete	Yes	Yes
Space Complexity	High	Low
Time Complexity	Depends on path cost	$O(b^d)$
Memory	Stores frontier	Very efficient

Advantages

- UCS always finds the least-cost path.
- IDS requires less memory.
- Both algorithms are complete under suitable conditions.

Disadvantages

- UCS consumes more memory.
- IDS repeats node expansions.
- Both may be slow in very large search spaces.

Relation to Intelligent Agents

The robot acts as a goal-based intelligent agent because its primary objective is to reach the exit. If the environment changes dynamically, a model-based agent becomes more suitable because it stores explored paths and updates its knowledge while moving.

Recommended Solution

A combination of a Model-Based Goal-Based Agent with Uniform Cost Search is recommended. The model-based component remembers explored regions, the goal-based

component keeps the robot focused on reaching the exit, and UCS guarantees the minimum-cost path.

Conclusion

Combining intelligent agent architectures with efficient search strategies enables autonomous robots to solve navigation problems accurately, safely, and efficiently even in unknown environments.